

ET 503: Electrical Machine and Drives in Electric Transportation

Course Code	ET 503
Course Name	Electrical Machine and Drives in Electric Transportation
L-T-P-C	3-0-0-3
Prerequisites	Electromechanics (EE 201), Control System (EE301) and Power Electronics (EE 309)
Intended for	UG/PG
Mutual Exclusion	None
Approval	44th BoA (Nov 2021)

Course Contents

- **Introduction to Electric Transportation:** Example of EVs; State of the art in Electric Vehicle Technology, Overview of EV technologies, Fuel Cell Electric Vehicles, Hybrid Electric Vehicles (HEVs), Vehicle Dynamics and Drive cycle, Introduction to Railway Systems. (5 Hours)
- **Fundamentals of Electric Drive control:** Control Block diagram, Reference frame Theory, dq-reference frame, PI- controllers Design, Hysteresis controller Control Methods: Field oriented control, Direct torque control, Sensor less control, Model Reference Adaptive Control (MRAC) Approach, Sliding mode control DC machine Drive: Closed loop Speed Control of DC Motor Through armature voltage control and field control, Regenerative Braking Control of motors in the EV: Multi Wheel Drive (MWD) or All Wheel Drive (AWD) systems, Torque vectoring etc. (12 hours)
- **Induction Motor Drives:** Induction Machines: Squirrel Cage Induction Machine and Slip Ring Induction Machines $\alpha\beta$ and dq-modeling of Induction Machines, Inverters for Induction Motors, PWM Switching Inverters, Soft-Switching Inverters Induction Motor Control: Voltage by Frequency Control, Field-Oriented Control, Direct Torque Control, Design Criteria of Induction Motor Drives for EVs, Design Example of Induction Motor Drives for EVs, Application Examples of Induction Motor Drives in EVs and Railways. (10 hours)
- **Permanent Magnet Brushless Motor Drives:** PM Materials, PM Brushless Machines: Structure of PM Brushless Machines, Principle of PM Brushless Machines, Modeling of PM Brushless Machines, Inverters for PM Brushless Motors, Inverter Requirements, Switching Schemes for Brushless AC Operation, Switching Schemes for Brushless DC Operation PM Brushless Motor Control: PM Synchronous Motor Control, PM Brushless DC Motor Control Design Criteria of PM Brushless Motor Drives for EVs: Design Examples of PM Brushless Motor Drives for EVs, Planetary-Geared PM Synchronous Motor Drive, Outer-Rotor PM Brushless DC Motor Drive, Application Examples of PM Brushless Motor Drives in EVs and Railways. (11 hours)
- **Introduction to Special Machines:** Switched Reluctance machine (SRM), Synchronous Reluctance Machine (SyRM) etc. (4 hours)

Laboratory Modules

Nil

Text Books

1. K. T. CHAU, **Electric Vehicle Machines and Drives: Design, Analysis and Application**, Wiley-IEEE Press, 2015.
2. Morris Brenna, Federica Foiadelli, Dario Zaninelli, **Electrical Railway Transportation Systems**, Wiley-IEEE Press, 2018.

References

1. Ned Mohan, Siddharth Raju, **Analysis and Control of Electric Drives: Simulations and Laboratory Implementation**, Aug. 2020
2. John G. Hayes, G. Abas Goodarzi, **Powertrain: Energy Systems, Power Electronics and Drives for Hybrid, Electric and Fuel Cell Vehicles**
3. W. Leonhard, **Control of Electric Drives**, 2001.
4. P. Vas, **Sensorless Vector and Direct Torque Control**, Oxford Science Publications
5. Boldea, S. A. Nasar, **Electric Drives**, 2nd Edition, CRC Press, 2006.
6. Bose B.K., **Power Electronics and Variable Frequency Drives – Technology and Applications**, IEEE Press, Standard Publisher Distributors. 2001
7. Rashid M., **Power Electronics- Circuits, Devices and Applications**, 3rd Edition, Pearson Education.
8. Krause, P. C., Wasynczuk, O., Sudhoff, S. D., **Analysis of Electric Machinery and Drive Systems**, Wiley-Interscience.
9. S. K. Pillai, **A First Course on Electrical Drives**, New Age International Pvt. Ltd.
10. R. Krishnan, **Electric Motor Drives: Modeling, Analysis, and Control**, Prentice Hall, 2001.